Lake Encroachment Watch Product Requirements Document

Case Study 6, Government / Public-Sector track.

Lake Encroachment Watch is an early-warning tool for Bengaluru's lake NGOs and preservation communities. When new construction appears inside a lake's legal buffer zone, it gives them a reviewable, evidenced case file they can confirm and act on — and it helps them file the RTI or complaint. The pilot covers three lakes: Bellandur, Varthur, and Agara.

Approach and scope

The problem was given and fixed by the case: the stakeholder (lake NGOs and communities), the bottleneck (finding buffer-zone construction early enough to act), and the credibility constraint. Discovery was therefore kept light, and the effort went into product thinking, a focused MVP, real analytics, and a deployed build that real people use.

This was an individual build with a group sync, on a roughly one-day sprint to a hard deadline. The end-of-sprint deliverable is a PRD plus a deployed MVP plus real users and real Mixpanel data — not a PRD alone, and not a PRD with a demo.

Once the problem, persona, and scope were locked at the start (one lake, one real image-diffed case, one reviewed case file), there were no major pivots. The scope was expanded once, deliberately — to three lakes, a working RTI/complaint dossier generator, and a complaint-tracking loop — to make the paid wedge real rather than a teaser. That change is documented here; the problem and persona did not change.

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1. The problem

Bengaluru's lakes are under continuous encroachment pressure. As the city's real-estate frontier pushes into low-lying tank beds and their margins, construction appears inside the legally protected buffer around lakes. The binding constraint is detection, not law: the buffer rule already exists; what is missing is anyone noticing a violation early enough to act. By the time an encroachment is visible on the ground, the structure is built, occupied, and defended in court, where reversal is slow, expensive, and often lost.

Stated plainly: NGO lake-volunteers and lake-preservation communities cannot tell, early enough to act, when new construction has appeared inside a lake's legal buffer zone.

Satellite imagery, RTI, and the courts all exist, yet the core outcome — catching a violation while it can still be stopped, with evidence that survives a hostile agency — still is not reached. There is also a trap: a single false alarm burns an NGO's credibility with the agencies it needs, so a naive "ALERT: illegal" tool is worse than nothing.

The question this product answers:

> ***What should the first version put in a lake-defender's hands so they can catch a buffer-zone encroachment early and stand behind it — and how will we know, once it is live, that real defenders find it credible and worth paying for?***

Out of scope, deliberately: automatic legal verdicts; city-wide or all-lake coverage; live imagery ingestion at scale; real email or WhatsApp delivery; authentication. These are roadmap items, covered in sections 5 and 10.

The first version's job, in one sentence: surface one buffer-zone change as a reviewable, evidenced case file that a human can confirm and act on. The wider opportunity — protecting 182-plus lakes across the Greater Bengaluru Area — is kept separate from the narrow bottleneck fixed here: early, credible detection at one lake.

2. Users and influence mapping

This is a community and NGO journey with a buyer-is-not-user split and several external influencers, not a multi-team B2B approval chain. The friction lives in the handoffs — detection to evidence to complaint to enforcement — and in a missing early nudge, not inside any one party.

Party	Cares most about	Where they help or block	Influence	Role
Lake volunteer / community member	Their lake surviving; not wasting effort	Ground-truths pins, raises alarms	Medium	Primary user
NGO / foundation (FBL, Paani.Earth, Friends of Lakes)	Credibility, impact, funding	Files complaints and RTIs, mobilises	High	Buyer and power user
NGO reviewer / core member	Not crying wolf	The credibility gate: confirm or flag	High	Primary user (reviewer)
BBMP / BDA (civic agencies)	Their own liability	Approve and act, or stonewall	High	Influencer / target
KTCDA / NGT / courts	Rule enforcement	Ultimate teeth; slow	High	Influencer / arbiter
RWAs / neighbourhood groups	Local amenity, property	Cheap per-lake subscribers	Medium	Secondary user / buyer

CSR / corporate lake-adopters	Visible impact, brand	Fund branded monitoring	Medium	Secondary buyer
Donors / grant-makers	Measurable outcomes	Fund the org	Medium	Influencer of buyer

The NGO or foundation is the buyer and will pay for dossiers and multi-lake monitoring. The volunteer and the NGO reviewer are the primary users. They are not the same party: the reviewer needs defensible evidence, and the volunteer needs to know where to go — different needs that drive different features.

The seam every existing approach ignores is the reviewer's credibility need. Raw satellite tools and citizen-report apps hand over unverified signals; nobody packages a change as pre-reviewed, evidenced, confidence-labelled material an NGO can put its name on. That is the wedge.

3. The current journey

Where time, trust, and momentum are lost between "a violation starts" and "it's stopped."

Stage	What happens / friction	Cost
1. Violation begins	Clearing or foundation starts inside the buffer. Nobody is watching that spot.	Silent
2. Chance discovery	A volunteer notices months later, by accident, usually once it is visible or built.	Weeks to months lost
3. Is it even a violation?	Is this inside the buffer? Which rule? No easy map. Argued internally.	Hours to days
4. Evidence gathering	Manually assemble imagery, dates, coordinates, records checks.	Days of manual work
5. Credibility gut-check	"Are we sure? If we're wrong, we lose standing." Often stalls here.	Drop-off point
6. Complaint / RTI	File with BBMP / BDA / NGT; frequently vanishes.	Days plus uncertainty
7. Enforcement	Slow; by now built and defended.	Often lost

The MVP targets stages 2 to 5, collapsed: getting from "a change happened" to "a credible, evidenced case we would put our name to," early. Stage 2 — late, accidental discovery — is the root cause; stages 4 and 5 — manual evidence and credibility fear — are where good intentions die.

The symptom is that encroachments get built. The root cause is that there is no continuous, credible early-detection layer, so defenders are always reacting late with hand-built, contestable evidence.

4. Secondary research

Lean, sourced framing rather than a market-sizing exercise.

What is known and real:

- The legal buffer exists. The National Green Tribunal's 2016 order established a 75 m no-development buffer around lakes; Karnataka's tank-conservation framework (KTCDA) administers lake protection and buffer enforcement. This is the legal spine the product overlays.
- Three real, contested lakes anchor the pilot: Bellandur (about 361 ha, chronically in the news for froth, fire, and encroachment), Varthur (about 180 ha, downstream, similar pressure), and Agara (about 40 ha, HSR / Sarjapur). All have active defender communities and span two acreage tiers.
- Free satellite imagery is viable. Sentinel-2 provides free imagery at about 10 m resolution with a roughly 5-day revisit — enough to detect structure-scale change, which makes an automated diff pipeline feasible at zero imagery cost.

Note: specific market-size figures — encroachments per year, rupee value of affected land, count of active lake NGOs — were not supplied by the case and are not asserted here; sizing is deferred rather than invented. The exact 2025 acreage-tiered buffer distances must be confirmed against the current gazette before any real complaint cites them. The MVP encodes illustrative tiered values: 75 m for Category-A lakes over 40 ha, and 60 m for Category-B lakes of 30 to 40 ha.

Both the law and the free imagery already exist. The missing piece is a product layer — detection, credibility, and workflow — not new legislation or expensive data. That is the wedge the MVP occupies.

5. Existing solutions, and the job they miss

Organised by the job to be done, not by brand.

Category	Does well	Where it stops
Raw satellite / GIS (Sentinel Hub, Google Earth)	Shows imagery; power users can compare	Doesn't overlay the legal buffer, doesn't flag change, doesn't package evidence — leaves all the work to the user
Citizen-report apps / WhatsApp groups	Crowd surfaces sightings	Unverified, uneven, late (post-construction); no credibility layer, so agencies dismiss it
RTI / complaint portals	A formal channel exists	Assume you already have the evidence pack; nothing helps you build it or catch the violation early

NGO manual monitoring	Deep local knowledge	Human, non-continuous, does not scale; evidence rebuilt by hand each time
One-off news / academic studies	Credible, detailed	Retrospective, slow, not an operational early-warning tool

The job none of them finishes: tell me, early and credibly, that a buffer-zone violation has appeared, and hand me evidence I can act on without a week of manual work. Each existing tool solves a different job — view imagery, collect reports, file complaints — so no one owns the early-plus-credible-plus-evidenced detection seam. That is the MVP's target.

6. Primary research

Because the problem is given, research is light and aimed at locating the bottleneck and validating the MVP, not at finding a new problem. A reasonable target is 6 to 10 conversations with real lake defenders, anchored on their last specific encroachment episode.

Note: no formal interview round was conducted within this one-day sprint, and no interview quotes or counts are fabricated. The problem, scope, and credibility constraint were locked through a structured problem-grilling session documented in the project plan. Primary validation runs as a live launch to real lake-group contacts (section 14), where actual behaviour — confirm, flag, subscribe, dossier-intent — is the signal. That is stronger than stated-preference interviews.

Planned research instrument, to run at or after launch (structure is illustrative; fill with real responses):

Segment (n=)	**Method**	**What to probe**
NGO reviewers, FBL / Paani.Earth / Friends of Lakes	Walk them through a live case file	"Would you put the org's name on this? What's missing?"
Lake volunteers	Field-test "Go verify this pin"	Did they reach the right spot?
Foundation lead / buyer	Show the dossier	"Would you pay for the finished dossier? How much?"

Interviewing discipline: anchor on the last real episode, not "usually"; keep talk-share low; ask open why, what, and how; never lead; and distrust "I'd use this" — read what people actually do in the product. A confirm_case or click_generate_dossier event beats a compliment.

Kill-criterion, decided in advance: if reviewers open case files but refuse to confirm or flag them (credibility not earned), or ignore the dossier action (no willingness to pay), the core hypothesis is wrong. That is what section 14's events watch for.

7. Opportunity and prioritisation

Candidate opportunities, then a commitment on value over effort.

- A. For NGO reviewers who fear crying wolf, through pre-reviewed, evidenced, confidence-labelled case files, because no tool owns the credibility seam.
- B. For volunteers who find out late, through a change feed plus subscribe per lake, because nobody watches continuously.
- C. For NGOs drowning in manual evidence work, through a one-click complaint/RTI dossier, because existing portals assume you already have the pack.
- D. For foundations funding many lakes, through a multi-lake monitoring dashboard, because monitoring does not scale by hand.

Candidate	Value	Effort	Carry forward?
A. Evidenced case file + confidence + confirm/flag	High	Medium	Yes — the core
B. Change feed + subscribe	Medium	Small	Yes — pairs with A
C. One-click dossier	High	Large	Validate demand first
D. Multi-lake dashboard	Medium	Large	Roadmap

Carried forward: A and B as the MVP — credible case files plus the browse-and-subscribe loop. A beats a feed alone because credibility, not awareness, is the binding constraint. C is the highest value but the highest effort, so the MVP measures its demand before it is built out. During the sprint, C was in fact built as a working generator (see section 10) once time allowed — the decision to make the paid wedge real rather than a stub.

8. Product strategy

Primary users: the NGO reviewer or core member (the credibility gate) and the lake volunteer (ground-truth). Buyer: the NGO or foundation, distinct from the volunteer user.

Persona — Asha, a core member of a Bengaluru lake NGO. She has fielded a dozen "I think someone's building on the lake" messages this year. The problem bites the day she has to decide whether to escalate one to BBMP, knowing a wrong call costs the org standing it cannot rebuild. Success for her is opening something she can forward to an agency today, without a week of prep, confident it will hold. What she wants: catch it early, and be believed.

Problem statement:

> ***For NGO lake-defenders and their volunteer communities, spotting new construction inside a lake's legal buffer early and credibly is the bottleneck, because there is no continuous detection layer that also packages a change as reviewable, defensible evidence. Late or contestable alarms let encroachments get built, defended, and normalised, while a single false alarm burns the credibility defenders depend on.***

Scope was locked at sprint start — the map, feed, evidenced case file, confirm/flag, and subscribe loop — and then expanded once, deliberately, to three lakes, a working RTI/complaint dossier generator, and the complaint-tracking loop. The problem and persona did not change.

9. Solution direction

Three distinct directions were considered:

Direction	Shape	User value	Effort	Risk
1. Alert firehose — auto-flag every change, push notifications	Speed-first	Medium	Medium	High — cries wolf, kills credibility
2. Map-first evidenced case files — human in the loop	Credibility-first	High	Medium	Low — matches the constraint
3. Report portal — crowd submits, we route	Crowd-first	Medium	Medium	Medium — unverified, late

Chosen: direction 2 — a map-first product where every detection is an evidenced, confidence-labelled case file gated by a human confirm or flag. The binding constraint is credibility, not raw speed or volume. Direction 1 destroys the value prop; direction 3 reproduces the late, unverified problem defenders already have. Direction 2 turns detection into something an NGO can put its name on.

End-to-end flow:

Step	Today	With the product
Notice	Accidental, months late	Map or feed surfaces the change
Judge	Argue internally	Open the case file: imagery, buffer distance, confidence
Trust	Gut fear	Confirm on ground or flag as noise (the credibility gate)
Act	Days of manual prep	Verify the pin on a phone, generate and file an RTI/complaint dossier, track it

Every feature in section 10 traces back to a step here.

10. Product detail and MVP scope

Features mapped to the flow above and classified by product value:

#	Feature	Flow step	Class	Status
F1	Multi-lake overview map and per-lake maps: boundary, acreage-tiered legal buffer, watch band, change pins	Notice	Must-have	Shipped (3 lakes)
F2	Developments feed (severity-ranked cards)	Notice	Must-have	Shipped
F3	Case file: before/after slider, real computed-diff overlay, bounding box	Judge	Must-have (the core)	Shipped
F4	Evidence block: imagery dates, changed percentage, buffer distance, records narrative, GPS	Judge	Must-have	Shipped
F5	Confidence label (high / medium / low)	Judge, Trust	Must-have	Shipped
F6	Confirm on ground / Looks like noise (credibility gate)	Trust	Must-have	Shipped
F7	Subscribe to a lake or all lakes	Notice (retain)	Performance	Shipped
F8	"Go verify this pin" — GPS on a phone	Act	Delighter	Shipped
F9	RTI / complaint dossier generator (RTI Act 2005 or complaint; addressee picker;	Act	Must-have (paid wedge)	Shipped

	evidence auto-filled; print, copy, and official-portal link)			
F10	Complaint tracker (Draft, Filed, Acknowledged, Resolved, plus reference number)	Act	Performance	Shipped
F11	Shareable WhatsApp evidence card	Notice, reach	Delighter	Shipped
F12	Per-lake change-history timeline	Judge, retain	Performance	Shipped
F13	Live Sentinel-2 imagery and auto change-detection (cloud-filtered, NDWI water-mask, weekly job, "Check now")	Notice	Must-have	Shipped (env-gated; runs on a stored pair without a key)
F14	Human-gated review queue — auto-detected changes become pending candidates, promoted or dismissed by a reviewer	Trust	Must-have	Shipped
F15	Satellite basemap toggle (Esri World Imagery)	Notice	Delighter	Shipped
F16	Shared database (Supabase) — reviews, candidate decisions, subscribers, complaints, dossier-intent	All	Must-have	Shipped (env-gated; localStorage fallback)

F17	Records cross-ref (BBMP / BDA) — automatic approval check	Judge	Performance	V1.1
F18	Direct auto-submission plus email/WhatsApp delivery	Act, Notice	Performance	V1.1
F19	Donor / CSR impact dashboard	Scale	Paid	V2

The shipped MVP is F1 through F16. Scope was expanded past the original lock — one lake and a teaser dossier — into a working three-lake product with a real dossier generator and the file-and-track loop, because the paid wedge and the "complaints vanish" pain are what make it something NGOs adopt and pay for.

Edge cases handled: when a change is diffuse and edge-less (seasonal), the product surfaces it at low confidence and states it is not an encroachment claim (Bellandur case 3) rather than crying wolf. When the map fails to load, the feed is a full fallback. When analytics or persistence is unavailable, events log to the console and localStorage. The dossier always carries a "review and edit before filing; confirm the exact gazetted buffer" disclaimer — the tool never claims legal certainty.

Information architecture: the home route (overview map and feed, plus lake cards) leads to a per-lake route (focused map, feed, timeline), which leads to a case route (evidence, review, act, dossier, tracker).

Deliberately not built, and why: live imagery ingestion at scale (the curated real diff proves the pipeline; scale is roadmap); automatic records cross-ref (BBMP / BDA data is patchy, so the narrative is curated); direct auto-submission to government portals (a legal filing must stay a human decision — the product hands over the filled document and the portal link); and authentication (open browsing maximises pilot reach).

11. UX and design

The design was worked in order: conceptual, then information, then interaction, then visual.

- Conceptual: the product is a case file, not an alarm. Every screen reinforces "evidence for review," never "verdict."
- Information: a map-first landing (the visual hook) with a feed alternative. The case file leads with imagery, then evidence, then the review gate, then the action — the order a skeptical reviewer thinks in.
- Interaction: a draggable before/after slider; a diff-overlay toggle that reveals the computed change and its bounding box; one-tap confirm or flag; verify-pin opens the phone's map.

- Visual: a restrained government/NGO palette. Red is reserved for in-buffer violations, amber for the watch band, cyan for low-confidence or noise. Dense but calm, and mobile-first, because volunteers are on phones.

Self-check against six components:

Component	How judged	This build
Usability	Complete the core task unaided	A two-screen loop, verified end-to-end in a real browser
Efficiency	Time to judge a case	Everything needed to decide is on one case-file screen
Perceived effort	Clutter or heaviness	Low — one accent colour with meaning, generous spacing
Credibility	Trust cues, no broken states	Dates, sources, a "reviewable finding, not a verdict" disclaimer, confidence bars
Delight	One unexpected moment	The diff-overlay reveal and "Go verify this pin" straight to GPS
Simplicity	Taps to outcome	Map, pin, case file, confirm — three taps

Reusable components built: severity and confidence badges, case card, evidence rows, before/after slider, review actions, and the dossier and subscribe forms — a small, consistent kit that matches the shipped code.

12. Analytics

Analytics is wired into the product, not attached as a slide.

Metrics:

- North Star: confirmed, acted-on case files — evidenced buffer-zone changes that a human confirmed and moved on. That is the delivered value: credible catches, early.
- Leading: case-file open rate (view_case_file over view_map); confirm/flag rate per case; verify-pin rate.
- Lagging: subscribe conversion; dossier-intent rate (willingness to pay); confirmed-to-flagged ratio (credibility health).
- Activation: the first confirm_case or click_generate_dossier — the moment a user trusts the evidence enough to act.

13. Build and deployment

Live link: https://lake-encroachment-watch-bengaluru.vercel.app

Stack:

- Frontend and SSR: Next.js 15 (App Router, TypeScript), deployed on Vercel.
- Map: react-leaflet and Leaflet over OpenStreetMap tiles (no API token). Three lake boundaries as GeoJSON; acreage-tiered buffers (60 and 75 m legal, plus watch bands) computed with @turf/turf.
- Detection: build-time scripts/gen-tiles.mjs (synthetic tiles) and scripts/diff.mjs (sharp and pngjs) compute a genuine pixel diff — changed percentage, bounding box, diff.json, and an orange diff-overlay PNG per case (6 cases).
- Dossier: client-side document assembly (RTI Act 2005 and complaint templates), with browser print-to-PDF, clipboard copy, and official-portal deep links.
- Analytics: mixpanel-browser via a typed wrapper, with console and localStorage fallback.
- Persistence: Supabase when configured, with localStorage as the per-browser fallback (reviews, subscribe, complaint status).

14. Launch and real-world signal

Section 13 stops at "deployed and tested." This case also requires real users and real Mixpanel data.

Channel, stated honestly: direct outreach to two or three real FBL / Paani.Earth / Friends-of-Lakes contacts and Bengaluru lake WhatsApp groups — a warm channel that is repeatable with effort, appropriate for a pilot, not a scalable growth loop. A repeatable loop (shareable case-file cards, per-lake subscribe) is roadmap.

- Note: real launch numbers are to be filled in at submission. Do not report projections as results.